

MTH2021 – Complex Analysis

Part 2

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1 Integration

We integrate functions along curves in the complex plane. It is convenient to define the curve parametrically:

$$z = z(t) = x(t) + iy(t)$$

where t is a real parameter. Then a curve C is the set of points $z(t)$ for $\alpha \leq t \leq \beta$ joining $z(\alpha)$ and $z(\beta)$.

If $x(t)$, $y(t)$ are continuously differentiable, with $z'(t) \neq 0$, the curve is smooth. We consider integration along piecewise smooth curves (a finite number of smooth curves joined end to end).

Consider a given function $f(z)$. We divide the curve into n sub-intervals by the points $z_0 = z(\alpha)$, $z_1, \dots, z_{n-1}$, $z_n = z(\beta)$.

Let $\Delta z_k = z_k - z_{k-1}$, and let $\bar{z}_k$ be a point on the curve in the sub-interval between z_{k-1} and z_k . Then the limit of the Riemann sum

$$\lim_{n \rightarrow \infty} \left[\sum_{k=1}^n f(\bar{z}_k) \Delta z_k \right]$$

defines the line integral of $f(z)$ along C . We write this integral as

$$\int_C f(z) dz$$

It can be evaluated in the same way as a real integral:

$$\int_C f(z) dz = \int_{\alpha}^{\beta} f(z(t)) \frac{dz(t)}{dt} dt$$

The integration goes hand in hand with a choice of direction along the curve. Reversing the direction introduces a minus sign:

$$\int_{C:P \rightarrow Q} f(z) dz = - \int_{C:Q \rightarrow P} f(z) dz$$

The integral can be split into sub-integrals by splitting the curve C into sub-intervals $C : P \rightarrow Q \rightarrow R$

$$\int_{C:P \rightarrow R} f(z) dz = \int_{C:P \rightarrow Q} f(z) dz + \int_{C:Q \rightarrow R} f(z) dz$$

A contour integral is an integral over a closed curve (also called a closed contour) and is written as

$$\oint_C f(z) dz$$

The positive direction is the anti-clockwise direction.

Examples

- $C : z(t) = 1 + it$ for $-1 \leq t \leq 1$ and $f(z) = z + 1$

$$\begin{aligned} \int_C f(z) dz &= \int_C f(z) \frac{dz(t)}{dt} dt = \int_{-1}^1 [z(t) + 1] i dt \\ &= \int_{-1}^1 (2 + it) i dt \\ &= i [2t + it^2/2]_{-1}^1 = 4i \end{aligned}$$

- $C : z(t) = t + it^2$ for $0 \leq t \leq 1$ and $f(z) = z^2$

$$\begin{aligned} \int_C f(z) dz &= \int_0^1 (t + it^2)^2 (1 + 2it) dt \\ &= \int_0^1 [t^2 - 5t^4 + i(4t^3 - 2t^5)] dt \\ &= [t^3/3 - t^5 + i(t^4 - t^6/3)]_0^1 \\ &= -2/3 + 2i/3 \end{aligned}$$

- $C : z(t) = r \exp(it)$ for $0 \leq t \leq 2\pi$ and $f(z) = 1/z^{n+1}$, $n = 0, 1, 2, \dots$

$$\begin{aligned} \oint_C f(z) dz &= \int_0^{2\pi} \left[r^{-(n+1)} \exp(-i(n+1)t) \right] [ir \exp(it)] dt \\ &= \frac{i}{r^n} \int_0^{2\pi} \exp(-int) dt \end{aligned}$$

For $n = 0$

$$\oint_C f(z) dz = i \int_0^{2\pi} dt = 2\pi i$$

For $n > 0$

$$\oint_C f(z) dz = \frac{i}{r^n} \int_0^{2\pi} (\cos nt - i \sin nt) dt = 0$$

This result will arise frequently from now on.

• C is the closed curve consisting of the line C_1 from $-R$ to R on the real axis and the semi-circle C_2 from R back to $-R$ in the upper half-plane

$$C_1 : z_1(t) = t \quad \text{for} \quad -R \leq t \leq R$$

$$C_2 : z_2(t) = R \exp(it) \quad \text{for} \quad 0 \leq t \leq \pi$$

$$f(z) = z^2$$

$$\int_{C_1} f(z) dz = \int_{-R}^R t^2 dt = \frac{2}{3} R^3$$

$$\begin{aligned} \int_{C_2} f(z) dz &= \int_0^\pi [R^2 \exp(2it)] [iR \exp(it)] dt \\ &= iR^3 \int_0^\pi \exp(3it) dt \\ &= iR^3 \frac{1}{3i} [\exp(3it)]_0^\pi \\ &= \frac{R^3}{3} [\exp(3i\pi) - 1] = -\frac{2R^3}{3} \end{aligned}$$

$$\oint_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz = 0$$

This result follows from the Cauchy theorem.

2 Cauchy theorem

Cauchy theorem

If $f(z)$ is analytic everywhere within and on a simple closed curve C , then

$$\oint_C f(z) dz = 0$$

Proof (optional)

There are two proofs of the theorem, the easier requiring an extra assumption. We sketch out the easier proof.

Let $f(z) = u(x, y) + iv(x, y)$. Then

$$\begin{aligned}\oint_C f(z) dz &= \oint_C (u + iv) (dx + i dy) \\ &= \oint_C (u dx - v dy) + i \oint_C (v dx + u dy)\end{aligned}$$

Next we apply Green's theorem* to each term. Let R be the region enclosed by C . Then

$$\begin{aligned}&\oint_C (u dx - v dy) + i \oint_C (v dx + u dy) \\ &= \int \int_R (-v_x - u_y) dx dy + i \int \int_R (u_x - v_y) dx dy \\ &= 0\end{aligned}$$

where for the last step we have used the Cauchy-Riemann equations.

Cauchy's theorem is a pivotal result in Complex Analysis.

*Green's theorem

Let C be a simple closed contour enclosing a region R . Let $P(x, y)$ and $Q(x, y)$ be two real functions. If P and Q , and their first-order partial derivatives, are continuous on and within C , then

$$\oint_C (-P dx + Q dy) = \int \int_R (Q_x + P_y) dx dy$$

Morera theorem

If $f(z)$ is continuous in a region R and if

$$\oint_C f(z) dz = 0$$

for every closed curve C in R , then $f(z)$ is analytic in R .

3 Path independence

Cauchy's theorem has a number of immediate consequences.

The line integral of $f(z)$ between two points P and Q is the same for any two paths from P to Q , provided $f(z)$ is analytic on, and throughout the region between, the two paths.

To show this, choose two paths C_1 and C_2 from P to Q . Consider the closed path $C = C_1 - C_2$ and apply Cauchy's theorem:

$$\begin{aligned}\oint_C f(z) dz &= \left[\int_{C_1} - \int_{C_2} \right] f(z) dz \\ &= 0 \\ \Rightarrow \int_{C_1} f(z) dz &= \int_{C_2} f(z) dz\end{aligned}$$

4 Contour deformation

If $f(z)$ is analytic in and on the boundary of the region between two simple closed curves C_1 and C_2 , then

$$\oint_{C_1} f(z) dz = \oint_{C_2} f(z) dz$$

with each integral taken in the same sense.

To show this, consider a closed curve $C = \gamma_1 + L_{12} + \gamma_2 + L_{21}$ (sketched in lectures) as follows.

- L_{12} is a straight line joining C_1 and C_2 .
- L_{21} is a straight line joining C_2 and C_1 , parallel to L_{12} with a small separation ϵ from it.

The combination L_{12} and L_{21} is called a cross-cut.

- γ_1 is C_1 with a section of length ϵ removed from it where the cross-cut meets it.
- Similarly γ_2 is C_2 with a section of length ϵ removed from it where the cross-cut meets it, except that γ_2 is traversed in the opposite direction.

By Cauchy's theorem

$$\begin{aligned} \oint_C f(z) dz &= \left[\int_{\gamma_1} + \int_{L_{12}} + \int_{\gamma_2} + \int_{L_{21}} \right] f(z) dz \\ &= 0 \end{aligned}$$

In the limit $\epsilon \rightarrow 0$, $L_{12} \rightarrow -L_{21}$ and

$$\left[\int_{L_{12}} + \int_{L_{21}} \right] f(z) dz = 0$$

Also $\gamma_1 \rightarrow C_1$ and $\gamma_2 \rightarrow -C_2$. Hence

$$\oint_{C_1} f(z) dz = \oint_{C_2} f(z) dz$$

Therefore we can deform one closed contour into another, provided $f(z)$ remains analytic everywhere between and on the two contours.

This is a key tool for a number of further results.

5 Darboux inequality

The Darboux inequality puts an upper bound on the magnitude of integrals:

$$\left| \int_C f(z) dz \right| \leq ML$$

where $|f(z)| \leq M$ on C (i.e. M is an upper bound on $|f(z)|$ for points z on C) and L is the length of the curve C .

To show this, we apply the extension of the triangle inequality

$$\left| \sum_{k=1}^n z_k \right| \leq \sum_{k=1}^n |z_k|$$

to the Riemann sum $\sum_{k=1}^n f(\bar{z}_k) \Delta z_k$:

$$\begin{aligned} \left| \sum_{k=1}^n f(\bar{z}_k) \Delta z_k \right| &\leq \sum_{k=1}^n |f(\bar{z}_k) \Delta z_k| \\ &= \sum_{k=1}^n |f(\bar{z}_k)| |\Delta z_k| \end{aligned}$$

As $n \rightarrow \infty$ this becomes

$$\begin{aligned} \left| \int_C f(z) dz \right| &\leq \int_C |f(z)| |dz| \\ &\leq M \int_C |dz| \\ &= ML \end{aligned}$$

since $|dz| = \sqrt{dx^2 + dy^2}$ is an element of length along the curve.

6 Cauchy integral formula

We have the following cornerstone result.

Cauchy integral formula

If $f(z)$ is analytic everywhere within and on a simple closed curve C and z_0 is any point inside C , then

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$$

with the integral taken in the positive sense.

Proof

Deform C to a circle C_ϵ of radius ϵ about z_0 . Since the integrand is analytic throughout the region $C \rightarrow C_\epsilon$:

$$\begin{aligned} \oint_C \frac{f(z)}{z - z_0} dz &= \oint_{C_\epsilon} \frac{f(z)}{z - z_0} dz \\ &= \oint_{C_\epsilon} \frac{f(z) - f(z_0)}{z - z_0} dz + \oint_{C_\epsilon} \frac{f(z_0)}{z - z_0} dz \end{aligned}$$

The first term vanishes as $\epsilon \rightarrow 0$ by the Darboux inequality: since $f(z)$ is differentiable $\left| \frac{f(z) - f(z_0)}{z - z_0} \right|$ remains bounded as $z \rightarrow z_0$, while the length of C_ϵ vanishes. The second term gives

$$\oint_{C_\epsilon} \frac{f(z_0)}{z - z_0} dz = f(z_0) \oint_{C_\epsilon} \frac{dz}{z - z_0} = 2\pi i f(z_0)$$

where we have used one of the examples at the start of Integration.

Example

$$I = \oint_C \frac{\exp z}{z^2 + 1} dz$$

where C is the unit circle $|z - z_0| = 1$ with centre at:

(a) $z_0 = i$ (b) $z_0 = -i$

(a) We aim to write the integrand as $\frac{f(z)}{z - z_0}$ where $f(z)$ is analytic within and on C , and z_0 is within C . We have

$$z^2 + 1 = (z + i)(z - i)$$

and

$$\frac{\exp z}{z^2 + 1} = \frac{\exp z}{(z + i)(z - i)} = \frac{\frac{\exp z}{z + i}}{z - i} = \frac{f(z)}{z - z_0}$$

where

$$f(z) = \frac{\exp z}{z + i} \quad \text{and} \quad z_0 = i$$

The Cauchy integral formula then gives

$$I = 2\pi i f(z_0) = 2\pi i \frac{\exp i}{2i} = \pi (\cos 1 + i \sin 1)$$

(b) Now

$$\frac{\exp z}{z^2 + 1} = \frac{\exp z}{(z + i)(z - i)} = \frac{\frac{\exp z}{z - i}}{z + i} = \frac{f(z)}{z - z_0}$$

where

$$f(z) = \frac{\exp z}{z - i} \quad \text{and} \quad z_0 = -i$$

Hence

$$I = 2\pi i f(z_0) = 2\pi i \frac{\exp(-i)}{-2i} = -\pi (\cos 1 - i \sin 1)$$

Cauchy integral formula for derivatives

The Cauchy integral formula has a remarkable generalization.

- Consider

$$\begin{aligned} & \frac{f(z_0 + \Delta z_0) - f(z_0)}{\Delta z_0} \\ &= \frac{1}{2\pi i \Delta z_0} \left[\oint_C \frac{f(z)}{z - z_0 - \Delta z_0} dz - \oint_C \frac{f(z)}{z - z_0} dz \right] \\ &= \frac{1}{2\pi i \Delta z_0} \oint_C f(z) \left[\frac{1}{z - z_0 - \Delta z_0} - \frac{1}{z - z_0} \right] dz \\ &= \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0 - \Delta z_0)(z - z_0)} dz \end{aligned}$$

In the limit $\Delta z_0 \rightarrow 0$, the LHS by definition becomes $f'(z_0)$. To evaluate the RHS we use

$$\lim_{\Delta z_0 \rightarrow 0} \oint_C = \oint_C \lim_{\Delta z_0 \rightarrow 0}$$

(This requires separate justification which we omit but it is worth looking up in books.)

This gives

$$f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz$$

- Similarly

$$f''(z_0) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^3} dz$$

- The general form of the Cauchy integral formula follows by induction: if $f(z)$ is analytic everywhere within and on a simple closed curve C and z_0 is any point inside C , then the derivatives of $f(z)$ at z_0 of all orders exist and are analytic, and

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz$$

The fact that an analytic function is differentiable infinitely many times is one of the most important results in Complex Analysis.

Example

$$I = \oint_C \frac{\sinh z}{(z+1)^4} dz$$

where C is any simple closed contour enclosing $z_0 = -1$.

Write

$$I = \oint_C \frac{f(z)}{(z-z_0)^{n+1}} dz$$

where

$$f(z) = \sinh z \quad , \quad z_0 = -1 \quad , \quad n = 3$$

Then the Cauchy integral formula gives

$$I = \frac{2\pi i}{n!} f^{(n)}(z_0) = \frac{2\pi i}{3!} f^{(3)}(z_0)$$

For $f(z) = \sinh z$ we have $f^{(3)}(z_0) = f'''(z_0) = \cosh z_0$ and hence

$$I = \frac{2\pi i}{3!} \cosh(-1) = \frac{\pi i}{3} \cosh 1$$

7 Cauchy inequality

The Cauchy inequality puts an upper bound on the magnitude of derivatives of analytic functions:

if $f(z)$ is analytic within and on a circle, C , of radius R with centre at z_0 , and $|f(z)| \leq M$ on C , then

$$\left| f^{(n)}(z_0) \right| \leq \frac{n! M}{R^n}$$

We can see this from the Cauchy integral formula:

$$\begin{aligned} \left| f^{(n)}(z_0) \right| &= \left| \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz \right| \\ &\leq \frac{n!}{2\pi} \oint_C \frac{|f(z)|}{|z - z_0|^{n+1}} |dz| \\ &\leq \frac{n!}{2\pi} \frac{M}{R^{n+1}} \oint_C |dz| \\ &= \frac{n!}{2\pi} \frac{M}{R^{n+1}} 2\pi R = \frac{n! M}{R^n} \end{aligned}$$

8 Taylor series

The Cauchy integral formula leads to another key result.

Taylor's theorem

If $f(z)$ is analytic for all z such that $|z - z_0| < R_0$ (i.e. within a circle of radius R_0 centered at z_0), then for all such z

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \cdots + \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n + \cdots$$

- The maximum radius, R , about z_0 for which the above power series - also called Taylor series - converges is called the radius of convergence.
- If the function is analytic for all z in the complex plane (i.e. if it is entire), then R is infinite.
- If $f(z)$ has one or more singularities, then R extends to the nearest singularity.

Converse

Suppose the series

$$S(z) = \sum_{n=0}^{\infty} c_n(z - z_0)^n$$

converges for some point $w \neq z_0$, and let $R_0 = |w - z_0|$.

Then the series is absolutely convergent for all z such that $|z - z_0| < R_0$.

Furthermore, for all such z the series $S(z)$ is an analytic function of z .

Notes

- These results drop out of a series of theorems which we will not cover in lectures.
- The proof of Taylor's theorem is a self-study problem.
- We will shortly come to a generalization of Taylor series (Laurent series) whose derivation is a generalization of that of Taylor series.

Examples

- We have seen the expansions in powers of z - i.e. the Taylor series about $z_0 = 0$, sometimes called Maclaurin series - for the entire functions $\exp z$, $\sin z$, $\cos z$, $\sinh z$, $\cosh z$.
- We have also seen Maclaurin series with a finite radius of convergence, such as that for $\log(1 + z)$.
- Consider the Maclaurin series

$$f(z) = \frac{1}{(1 - z)^2} = 1 + 2z + 3z^2 + \cdots + (n + 1)z^n + \cdots$$

The Ratio test shows that it is absolutely convergent for all $|z| < 1$.

Now think of it as a Taylor series about $z_0 = 0$. The function $f(z)$ is singular at $z = 1$: this is the nearest singularity to the expansion point. Thus we should expect a radius of convergence $R = 1$, as is the case.

- Consider $f(z) = \exp(z + i)$.

Expansion about $z_0 = -i$:

$$\exp(z + i) = \sum_{n=0}^{\infty} \frac{(z + i)^n}{n!}$$

Expansion about $z_0 = 0$:

$$\begin{aligned} \exp(z + i) &= \exp i \exp z = \sum_{n=0}^{\infty} \frac{\exp i}{n!} z^n \\ &= \sum_{n=0}^{\infty} \frac{\cos 1 + i \sin 1}{n!} z^n \end{aligned}$$

Both converge absolutely for all z , consistent with the fact that $f(z)$ is entire.

9 Liouville theorem

Liouville theorem

If $f(z)$ is bounded and analytic for all values of z (bounded and entire), then $f(z)$ is constant.

Consider the Taylor series about $z = 0$ which converges for all z :

$$f(z) = f(0) + f'(0)z + \cdots + \frac{f^{(n)}(0)}{n!}z^n + \cdots$$

If $f(z)$ is bounded, then there exists M such that $|f(z)| \leq M$. Let C be a circle of radius R about $z = 0$. Then by the Cauchy inequality

$$|f^{(n)}(0)| \leq \frac{n! M}{R^n}$$

For any n , we can make this as small as we wish by allowing $R \rightarrow \infty$. Therefore all derivatives vanish, and $f(z) = f(0) = \text{const}$.

Fundamental Theorem of Algebra (optional)

Let $P(z)$ be a non-constant polynomial of finite degree. Suppose it has no zeros (i.e. it does not vanish for any finite z). Then

- i. $f(z) = \frac{1}{P(z)}$ is analytic everywhere
- ii. $\lim_{|z| \rightarrow \infty} |f(z)| = 0$, and therefore $f(z)$ is bounded.

But then by the Liouville theorem $f(z) = \text{const}$ and hence $P(z) = \text{const}$, which is a contradiction. Therefore a non-constant polynomial has at least one zero.

This is the Fundamental Theorem of Algebra.

10 Laurent series

Laurent series is a generalization of Taylor series for functions with singularities. Laurent expansions enable a major further development - the calculus of residues.

If $f(z)$ is analytic throughout an annulus centered at z_0 , then at any point z in the annulus $f(z)$ can be represented by the series

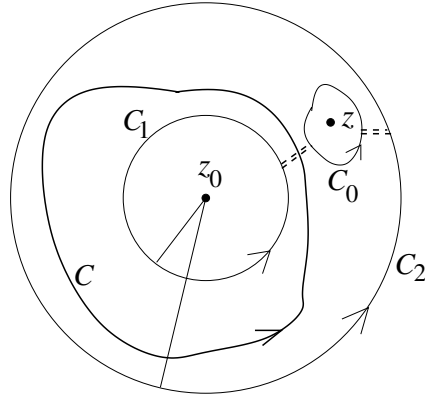
$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n \quad (\text{a})$$

where

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - z_0)^{n+1}} dw$$

with C being any simple closed contour around z_0 , lying within the annulus, taken in the positive sense. The Laurent series is unique.

Proof (optional)



Consider first a contour, C_0 , which encloses z and lies within the annulus. From the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i} \oint_{C_0} \frac{f(w)}{w - z} dw$$

Next we deform C_0 into $C_2 - C_1$ (using two cross cuts):

$$f(z) = -\frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{w - z} dw + \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{w - z} dw \quad (\text{b})$$

Consider term 1:

$$\begin{aligned}
-\frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{w-z} dw &= -\frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(w-z_0)-(z-z_0)} dw \\
&= \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(z-z_0) \left(1 - \frac{w-z_0}{z-z_0}\right)} dw \\
&= \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{z-z_0} \left(1 + \frac{w-z_0}{z-z_0} + \dots + \left(\frac{w-z_0}{z-z_0}\right)^n + \dots\right) dw
\end{aligned}$$

The series is absolutely convergent because for w on C_1 $\left|\frac{w-z_0}{z-z_0}\right| < 1$. With a bit of care we can interchange the summation and the integration:

$$\begin{aligned}
-\frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{w-z} dw &= \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{z-z_0} dw + \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)(w-z_0)}{(z-z_0)^2} dw + \dots \\
&= \sum_{n=-\infty}^{-1} (z-z_0)^n \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(w-z_0)^{n+1}} dw + \dots \\
&= \sum_{n=-\infty}^{-1} (z-z_0)^n \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w-z_0)^{n+1}} dw + \dots
\end{aligned}$$

where for the last step we have deformed C_1 back into a general simple closed contour C around z_0 , lying in the annulus.

This gives the negative powers of $(z-z_0)$ in the Laurent series in equation (a).

Next we consider term 2 in (b):

$$\begin{aligned}
\frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{w-z} dw &= \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{(w-z_0)-(z-z_0)} dw \\
&= \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{(w-z_0) \left(1 - \frac{z-z_0}{w-z_0}\right)} dw \\
&= \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{w-z_0} \left(1 + \frac{z-z_0}{w-z_0} + \dots + \left(\frac{z-z_0}{w-z_0}\right)^n + \dots\right) dw
\end{aligned}$$

The series is absolutely convergent because for w on C_2 $\left|\frac{z-z_0}{w-z_0}\right| < 1$. With a bit of care we can interchange the summation and the integration:

$$\begin{aligned}
\frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{w-z} dw &= \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{w-z_0} dw + \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)(z-z_0)}{(w-z_0)^2} dw + \dots \\
&= \sum_{n=0}^{\infty} (z-z_0)^n \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{(w-z_0)^{n+1}} dw + \dots \\
&= \sum_{n=0}^{\infty} (z-z_0)^n \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w-z_0)^{n+1}} dw + \dots
\end{aligned}$$

where for the last step we have deformed C_2 back into a general simple closed contour C around z_0 , lying in the annulus.

This gives the non-negative powers of $(z-z_0)$ in the Laurent series in equation (a).

Example

Find the Laurent series for

$$f(z) = \frac{1}{z^3(1-z)}$$

in the regions

(a) $0 < |z| < 1$

(b) $0 < |z - 1| < 1$

(c) $|z| > 1$

- Region (a) is within the unit circle about $z = 0$ excluding the centre.
- Region (b) is within the unit circle about $z = 1$ excluding the centre.
- Region (c) is outside the unit circle about $z = 0$.
- $f(z)$ has singularities at $z = 0$ and $z = 1$.

When working out Laurent series involves a binomial expansion, we aim to identify a quantity, in each region of interest, with a modulus < 1 , and expand in powers of it.

In region (a) $z_0 = 0$ and $|z| < 1$ so

$$\begin{aligned} f(z) = \frac{1}{z^3(1-z)} &= \frac{1}{z^3} \frac{1}{1-z} \\ &= \frac{1}{z^3} (1 + z + z^2 \dots) \\ &= \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{z} + 1 + z \dots \end{aligned}$$

This is the desired Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

in this region.

In region **(b)** $z_0 = 1$ and $|z - 1| < 1$. Let $w = z - 1$ and expand in powers of w :

$$\begin{aligned}
 f(z) = \frac{1}{z^3(1-z)} &= -\frac{1}{w} \frac{1}{(1+w)^3} \\
 &= -\frac{1}{w} \left(1 - 3w + \frac{(-3)(-4)}{2!} w^2 \dots \right) \\
 &= -\frac{1}{w} (1 - 3w + 6w^2 - 10w^3 \dots) \\
 &= -\frac{1}{w} + 3 - 6w + 10w^2 \dots \\
 &= a_{-1} (z-1)^{-1} + a_0 + a_1 (z-1) + a_2 (z-1)^2 \dots
 \end{aligned}$$

Notice how dramatically the series changes when we change the region. Having found the coefficients a_n we can use the earlier result

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - z_0)^{n+1}} dw$$

to evaluate contour integrals, for closed contours C within the given region. Of special interest is

$$a_{-1} = \frac{1}{2\pi i} \oint_C f(w) dw \quad \Rightarrow \quad \oint_C f(w) dw = 2\pi i a_{-1}$$

In the present case $a_{-1} = -1$ and hence

$$\oint_C \frac{1}{z^3(1-z)} dz = -2\pi i$$

In region **(c)** $z_0 = 0$ and $1/|z| < 1$. Let $w = 1/z$ and expand in powers of w :

$$\begin{aligned}
 f(z) = \frac{1}{z^3(1-z)} &= w^3 \frac{1}{1 - \frac{1}{w}} \\
 &= \frac{w^4}{w - 1} \\
 &= -\frac{w^4}{1 - w} \\
 &= -w^4 (1 + w + w^2 \dots) \\
 &= -w^4 - w^5 - w^6 \dots \\
 &= -\frac{1}{z^4} - \frac{1}{z^5} - \frac{1}{z^6} \dots
 \end{aligned}$$

(This is an expansion about $w = 0$: the point at infinity in terms of z .)

11 Singular points

- A singular point z_0 of a function $f(z)$ is a point where $f(z)$ is not analytic.
- It is an isolated singular point if $f(z)$ is analytic in a neighbourhood surrounding the point. Then $f(z)$ has a Laurent series about z_0

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

which is valid as far as the next singular point.

- If the negative powers in the Laurent series are finite in number:

$$f(z) = \frac{a_{-m}}{(z - z_0)^m} + \frac{a_{-m+1}}{(z - z_0)^{m-1}} + \dots$$

then $f(z)$ is said to have a pole of order m at z_0 .

- A pole of order 1 is sometimes called a simple pole. A pole of order 2 is sometimes called a double pole.
- If the number of negative powers is infinite, then $f(z)$ is said to have an essential singularity at z_0 .

Examples

i.
$$f(z) = \frac{12}{z^2} + \frac{3}{z} + 5z^2 + 7z^4 + \dots$$

has a pole of order 2 at $z = 0$.

ii.
$$f(z) = \frac{1}{z(z-1)^2}$$

has singularities at $z = 0$ (simple pole) and $z = 1$ (double pole).

iii.
$$\begin{aligned} f(z) &= \exp(z^{-1}) \\ &= 1 + \frac{1}{z} + \frac{1}{2!} \frac{1}{z^2} + \dots \end{aligned}$$

has an essential singularity at $z = 0$.

12 Cauchy residue theorem

- Suppose $f(z)$ has a pole of order m at z_0 . Then its Laurent series about z_0 reads

$$f(z) = \frac{a_{-m}}{(z - z_0)^m} + \cdots + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots$$

We have seen that

$$a_{-1} = \frac{1}{2\pi i} \oint_C f(z) dz$$

where C is a positively oriented simple closed contour about z_0 .

The coefficient a_{-1} is called the residue of $f(z)$ at z_0 .

The result

$$\oint_C f(z) dz = 2\pi i a_{-1}$$

is called the residue formula.

- If a simple closed curve C encloses several isolated singularities, then we can deform C into a set of circles, $C_1, C_2, \dots$, with circle C_i surrounding the singularity at z_i .

For example, for two singularities

$$\oint_C f(z) dz \rightarrow \oint_{C_1} f(z) dz + \oint_{C_2} f(z) dz$$

Applying the residue formula to each term on the right now gives

$$\oint_C f(z) dz = 2\pi i (r_1 + r_2)$$

where r_i is the residue of $f(z)$ at the singular point z_i .

This extends to any finite number of singularities.

Cauchy residue theorem

If $f(z)$ is analytic everywhere within and on a simple closed curve C , except at a finite number of singular points inside C , then

$$\oint_C f(z) dz = 2\pi i \sum \left(\begin{array}{c} \text{residues of } f(z) \\ \text{within } C \end{array} \right)$$

with the integral taken in the positive sense.

13 Evaluating residues

There are various ways of working out residues. With practice one can sometimes do it “by eye”. A formal procedure is as follows.

- If $f(z)$ has a simple pole (order 1) at z_0 , then

$$f(z) = \frac{a_{-1}}{z - z_0} + a_0 + a_1 (z - z_0) + \dots$$

Multiply by $(z - z_0)$:

$$(z - z_0) f(z) = a_{-1} + a_0 (z - z_0) + a_1 (z - z_0)^2 + \dots$$

Allow $z \rightarrow z_0$, and the residue is

$$a_{-1} = \lim_{z \rightarrow z_0} [(z - z_0) f(z)]$$

- If $f(z)$ has a pole of order 2 at z_0 , then

$$f(z) = \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1 (z - z_0) + \dots$$

Multiply by $(z - z_0)^2$:

$$(z - z_0)^2 f(z) = a_{-2} + a_{-1} (z - z_0) + a_0 (z - z_0)^2 + a_1 (z - z_0)^3 + \dots$$

Differentiate:

$$\frac{d}{dz} [(z - z_0)^2 f(z)] = a_{-1} + 2a_0 (z - z_0) + 3a_1 (z - z_0)^2 + \dots$$

Allow $z \rightarrow z_0$:

$$a_{-1} = \lim_{z \rightarrow z_0} \frac{d}{dz} [(z - z_0)^2 f(z)]$$

- If $f(z)$ has a pole of order m at z_0 , then

$$f(z) = \frac{a_{-m}}{(z - z_0)^m} + \cdots + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots$$

Multiply by $(z - z_0)^m$:

$$\begin{aligned} (z - z_0)^m f(z) &= a_{-m} + \cdots + a_{-2}(z - z_0)^{m-2} \\ &\quad + a_{-1}(z - z_0)^{m-1} + a_0(z - z_0)^m + \cdots \end{aligned}$$

Differentiate $(m - 1)$ times. This removes the coefficients $a_{-m}, \dots, a_{-2}$ on the right leaving

$$(m - 1)(m - 2) \dots (2)(1) a_{-1} + \text{positive powers of } (z - z_0)$$

Allow $z \rightarrow z_0$. This removes the positive powers leaving

$$\lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)] = (m - 1)! a_{-1}$$

Hence the residue is

$$a_{-1} = \frac{1}{(m - 1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]$$

14 Singularities: examples

We have defined poles in terms of the Laurent series. Working out the Laurent expansion about a given singular point is indeed one way of classifying a singularity and finding the residue (the coefficient a_{-1}). But we can often do so by other means.

Simplest case

$$f(z) = \frac{g(z)}{(z - z_0)^m}$$

where $g(z)$ is analytic at z_0 and $g(z_0) \neq 0$. Then $f(z)$ has a pole of order m at z_0 .

To show this, expand $g(z)$ in a Taylor series about z_0 :

$$\begin{aligned} f(z) &= \frac{g(z)}{(z - z_0)^m} \\ &= \frac{g(z_0) + g'(z_0)(z - z_0) + \frac{g''(z_0)}{2!}(z - z_0)^2 + \dots}{(z - z_0)^m} \\ &= \frac{g(z_0)}{(z - z_0)^m} + \frac{g'(z_0)}{(z - z_0)^{m-1}} + \dots \end{aligned}$$

But this is the Laurent series for $f(z)$ about z_0 . Since by assumption $g(z_0) \neq 0$, the first term is non-zero, the highest negative power of $(z - z_0)$ is m , and so $f(z)$ has a pole of order m at z_0 .

Example

$$f(z) = \frac{\exp z}{z(z - i)^3}$$

• Poles

The denominator vanishes at $z_1 = 0$ and $z_2 = i$, and each of these points is a candidate for a singularity.

For z close to $z_1 = 0$ we write

$$f(z) = \frac{\frac{\exp z}{(z-i)^3}}{z} = \frac{g(z)}{z} \quad \text{where} \quad g(z) = \frac{\exp z}{(z - i)^3}$$

$g(z)$ is analytic and non-zero at $z = 0$, while in the denominator z appears to the first power. Hence $f(z)$ has a pole of order 1 at $z = 0$.

For z close to $z_2 = i$:

$$f(z) = \frac{\frac{\exp z}{z}}{(z - i)^3} = \frac{g(z)}{(z - i)^3} \quad \text{where} \quad g(z) = \frac{\exp z}{z}$$

$g(z)$ is analytic and non-zero at $z = i$, while in the denominator $(z - i)$ appears to the third power. Hence $f(z)$ has a pole of order 3 at $z = i$.

• Residues

Having found the poles of

$$f(z) = \frac{\exp z}{z(z - i)^3}$$

we can evaluate the residue at each pole.

At $z_1 = 0$ (a pole of order 1) the residue is

$$r_1 = \lim_{z \rightarrow z_1} [(z - z_1) f(z)] = \lim_{z \rightarrow 0} \frac{\exp z}{(z - i)^3} = -\frac{1}{i^3} = \frac{1}{i} = -i$$

At $z_2 = i$ (a pole of order 3) the residue is

$$\begin{aligned} r_2 &= \frac{1}{2!} \lim_{z \rightarrow z_2} \frac{d^2}{dz^2} [(z - z_2)^3 f(z)] = \frac{1}{2!} \lim_{z \rightarrow i} \frac{d^2}{dz^2} \left(\frac{\exp z}{z} \right) \\ &= \frac{1}{2!} \lim_{z \rightarrow i} \left[\left(\frac{1}{z} - \frac{2}{z^2} + \frac{2}{z^3} \right) \exp z \right] \\ &= \frac{2 + i}{2} \exp i \\ &= \frac{2 + i}{2} (\cos 1 + i \sin 1) \end{aligned}$$

• Cauchy residue theorem

Consider

$$I = \oint_C f(z) dz = \oint_C \frac{\exp z}{z(z - i)^3} dz$$

If C does not enclose either z_1 or z_2 , then $I = 0$.

If C encloses z_1 , then $I = 2\pi i r_1 = 2\pi$.

If C encloses both z_1 and z_2 , then

$$I = 2\pi i (r_1 + r_2) = 2\pi i \left(-i + \frac{2 + i}{2} (\cos 1 + i \sin 1) \right)$$

Etc.

More complicated situations

Consider now

$$f(z) = \frac{g(z)}{(z - z_0)^m}$$

where $g(z)$ itself vanishes at $z = z_0$: $g(z_0) = 0$. Then whether $f(z)$ has a pole - and of what order - depends on the nature of the zero in $g(z_0)$.

- $$f(z) = \frac{\exp(z^2) - 1}{z}$$

Expanding $\exp(z^2)$ in a power series:

$$\begin{aligned} f(z) = \frac{\exp(z^2) - 1}{z} &= \frac{\left(1 + z^2 + \frac{z^4}{2!} + \dots\right) - 1}{z} \\ &= \frac{z^2 + \frac{z^4}{2!} + \dots}{z} \\ &= z + \frac{z^3}{2!} + \dots \end{aligned}$$

This is another convergent power series. Thus $f(z)$ is in fact analytic at $z = 0$ (and vanishes at $z = 0$: $f(0) = 0$).

- $$\begin{aligned} f(z) = \frac{\exp(z^2) - 1}{z^2} &= \frac{z^2 + \frac{z^4}{2!} + \dots}{z^2} \\ &= 1 + \frac{z^2}{2!} + \dots \end{aligned}$$

This again is a convergent power series; $f(z)$ is again analytic at $z = 0$ (though now $f(0) = 1$).

Removable singularities

To be precise, in the above examples $f(z)$ as written initially is undefined at $z = 0$. However the behaviour of $f(z)$ near $z = 0$ makes it possible to then define $f(0)$ so as to make it analytic, by setting $f(0) = \lim_{z \rightarrow 0} f(z)$. Such cases are called removable singularities.

$$\bullet \quad f(z) = \frac{\exp(z^2) - 1}{z^4} = \frac{z^2 + \frac{z^4}{2!} + \dots}{z^4} \\ = \frac{1}{z^2} + \frac{1}{2!} + \dots$$

This is a Laurent series with a highest negative power 2: now $f(z)$ has a pole of order 2 at $z = 0$.

General case

$$f(z) = \frac{g(z)}{h(z)}$$

where both $g(z)$ and $h(z)$ are analytic at some z_0 , and $h(z_0) = 0$.

To determine whether - and what - the pole is, we need to expand both functions in their respective power series about z_0 :

$$f(z) = \frac{a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \dots}{b_1(z - z_0) + b_2(z - z_0)^2 + \dots}$$

where in the denominator we have used $b_0 = h(z_0) = 0$, although there may be other vanishing coefficients too.

The nature of the singularity then depends on what the first non-zero coefficient is, in the numerator and in the denominator, and what is left behind after cancelling any common powers of $(z - z_0)$.

15 Evaluation of real integrals I

One of the most beautiful applications of Complex Analysis is the evaluation of classes of real integrals. We consider the most frequent types.

The first are integrals of the form

$$I = \int_0^{2\pi} F(\sin \theta, \cos \theta) d\theta$$

where F is a rational function, which is non-singular for $0 \leq \theta \leq 2\pi$.

We convert I to a contour integral and evaluate it using the residue theorem as follows.

- Put

$$\begin{aligned} z &= \exp(i\theta) \\ dz &= i \exp(i\theta) d\theta \quad \Rightarrow \quad d\theta = \frac{1}{i} \frac{dz}{\exp(i\theta)} = \frac{1}{i} \frac{dz}{z} \end{aligned}$$

As θ runs from 0 to 2π , z runs around the contour $|z| = 1$ once. This contour is the unit circle centered at the origin.

- Next:

$$\begin{aligned} \cos \theta &= \frac{\exp(i\theta) + \exp(-i\theta)}{2} = \frac{z + \frac{1}{z}}{2} \\ \sin \theta &= \frac{\exp(i\theta) - \exp(-i\theta)}{2i} = \frac{z - \frac{1}{z}}{2i} \end{aligned}$$

Then $F(\sin \theta, \cos \theta)$ becomes a function of z :

$$F(\sin \theta, \cos \theta) = F\left(\frac{z - \frac{1}{z}}{2i}, \frac{z + \frac{1}{z}}{2}\right) \equiv f(z)$$

$f(z)$ is a rational function, if F is.

- Finally the integral becomes

$$\begin{aligned} I = \int_0^{2\pi} F(\sin \theta, \cos \theta) d\theta &= \frac{1}{i} \oint_{|z|=1} \frac{f(z)}{z} dz \\ &= \frac{1}{i} 2\pi i \sum \left(\begin{array}{c} \text{residues of } \left[\frac{f(z)}{z} \right] \\ \text{within the unit circle} \end{array} \right) \end{aligned}$$

by the Cauchy residue theorem.

Example

The archetypal example is

$$I = \int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} \quad \text{with} \quad a > |b| > 0$$

We have

$$\begin{aligned} \int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} &= \frac{1}{i} \oint_{|z|=1} \frac{dz}{z} \frac{1}{a + \frac{b}{2} \left(z + \frac{1}{z}\right)} \\ &= \frac{2}{ib} \oint_{|z|=1} \frac{dz}{z} \frac{1}{\frac{2a}{b} + z + \frac{1}{z}} \\ &= \frac{2}{ib} \oint_{|z|=1} \frac{dz}{z^2 + 2\frac{a}{b}z + 1} \\ &= \frac{2}{ib} \oint_{|z|=1} \frac{dz}{(z - z_1)(z - z_2)} \end{aligned}$$

where

$$z_1 = \frac{-a + \sqrt{a^2 - b^2}}{b} \quad \text{and} \quad z_2 = \frac{-a - \sqrt{a^2 - b^2}}{b}$$

Since $z_1 z_2 = 1$ and $z_1 \neq z_2$, one of the roots is less than 1 in absolute value, and the other - greater than 1.

Indeed $|z_2| > 1$ since $a > |b|$. Hence $|z_1| < 1$.

Thus the simple pole at z_1 is inside the contour of integration, and z_2 - outside.

The residue at z_1 is

$$r_1 = \frac{2}{ib} \frac{1}{z_1 - z_2} = \frac{2}{ib} \frac{1}{2 \frac{\sqrt{a^2 - b^2}}{b}} = \frac{1}{i} \frac{1}{\sqrt{a^2 - b^2}}$$

By the residue theorem:

$$I = 2\pi i r_1 = \frac{2\pi}{\sqrt{a^2 - b^2}}$$

16 Real integrals II

The next type of integral is

$$I = \int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} dx$$

where $P(x)$ and $Q(x)$ are polynomials. We require the conditions:

- i. $Q(x) \neq 0$ for all real x
- ii. the degree of $Q(x)$ is at least 2 greater than the degree of $P(x)$

Then

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} dx = 2\pi i \sum \left(\begin{array}{c} \text{residues of } \left[\frac{P(z)}{Q(z)} \right] \\ \text{in upper half-plane} \end{array} \right)$$

Derivation (optional)

We integrate

$$f(z) = \frac{P(z)}{Q(z)}$$

around a closed contour, C , which runs from $-R$ to R along the real axis, and then from R back to $-R$ along the semi-circle, C_R , of radius R and centre at the origin, in the upper half-plane:

$$\oint_C f(z) dz = \int_{-R}^R f(x) dx + \int_{C_R} f(z) dz$$

Now we take $R \rightarrow \infty$ and apply the residue theorem:

$$\begin{aligned} \oint_C f(z) dz &= 2\pi i \sum \left(\begin{array}{c} \text{residues of } f(z) \\ \text{in upper half-plane} \end{array} \right) \\ &= \int_{-\infty}^{\infty} f(x) dx + \lim_{R \rightarrow \infty} \int_{C_R} f(z) dz \end{aligned}$$

The key is that the integral over C_R vanishes as $R \rightarrow \infty$ provided condition **ii.** is satisfied. On C_R : $z = R \exp(i\theta)$, $0 \leq \theta \leq \pi$ and

$$\begin{aligned} |f(z)| &= \left| \frac{P(z)}{Q(z)} \right| = \left| \frac{p_n z^n + p_{n-1} z^{n-1} + \dots}{q_k z^k + q_{k-1} z^{k-1} + \dots} \right| \\ &= \left| \frac{z^n}{z^k} \right| \left| \frac{p_n + p_{n-1} z^{-1} + \dots}{q_k + q_{k-1} z^{-1} + \dots} \right| \\ &= \frac{1}{R^{k-n}} \left| \frac{p_n + p_{n-1} R^{-1} \exp(-i\theta) + \dots}{q_k + q_{k-1} R^{-1} \exp(-i\theta) + \dots} \right| \end{aligned}$$

When $R \rightarrow \infty$ this dies out as $1/R^{k-n}$, while the length of C_R grows as πR . Then for $k - n \geq 2$ by the Darboux inequality $\left| \int_{C_R} f(z) dz \right| \rightarrow 0$.

This leaves

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} dx = 2\pi i \sum \left(\begin{array}{c} \text{residues of } \left[\frac{P(z)}{Q(z)} \right] \\ \text{in upper half-plane} \end{array} \right)$$

Example 1

$$I = \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^3}$$

Both conditions are satisfied.

Using $z^2 + 1 = (z + i)(z - i)$, we need the residues of

$$f(z) = \frac{1}{(1+z^2)^3} = \frac{1}{(z+i)^3(z-i)^3}$$

The singularities are at $z = \pm i$ and only $z = i$ is in the upper half-plane. This is a pole of order 3.

The residue is

$$\begin{aligned} r &= \frac{1}{2!} \lim_{z \rightarrow i} \frac{d^2}{dz^2} [(z-i)^3 f(z)] \\ &= \frac{1}{2!} \lim_{z \rightarrow i} \frac{d^2}{dz^2} \left[\frac{1}{(z+i)^3} \right] \\ &= \frac{1}{2!} \lim_{z \rightarrow i} \frac{12}{(z+i)^5} \\ &= \frac{3}{16i} \end{aligned}$$

Therefore

$$I = 2\pi i r = \frac{3\pi}{8}$$

Example 2

$$I = \int_{-\infty}^{\infty} \frac{x^2}{(x^2 + a^2)(x^2 + b^2)} dx \quad a > 0, b > 0$$

Both conditions are satisfied.

We need the residues of

$$f(z) = \frac{z^2}{(z^2 + a^2)(z^2 + b^2)} = \frac{z^2}{(z + ai)(z - ai)(z + bi)(z - bi)}$$

The singularities are at $z = \pm ai$ and $z = \pm bi$. In the upper half-plane we have simple poles at $z = ai$ and $z = bi$.

The residue at $z = ai$ is

$$\begin{aligned} r_1 &= \lim_{z \rightarrow ai} [(z - ai) f(z)] \\ &= \lim_{z \rightarrow ai} \left[\frac{z^2}{(z + ai)(z + bi)(z - bi)} \right] \\ &= \frac{a}{2i(a^2 - b^2)} \end{aligned}$$

Similarly the residue at $z = bi$ is $r_2 = \frac{b}{2i(b^2 - a^2)}$.

Therefore

$$I = 2\pi i (r_1 + r_2) = \frac{\pi}{a + b}$$

17 Real integrals III

Next we consider integrals of the form

$$I = \int_{-\infty}^{\infty} \exp(i\lambda x) \frac{P(x)}{Q(x)} dx$$

where $P(x)$ and $Q(x)$ are polynomials. We require the conditions:

- i. $Q(x) \neq 0$ for all real x
- ii. the degree of $Q(x)$ is at least 1 greater than the degree of $P(x)$ (note the difference compared with type II)
- iii. $\lambda > 0$

Then

$$I = \int_{-\infty}^{\infty} \exp(i\lambda x) \frac{P(x)}{Q(x)} dx = 2\pi i \sum \left(\begin{array}{c} \text{residues of } \left[\exp(i\lambda z) \frac{P(z)}{Q(z)} \right] \\ \text{in upper half-plane} \end{array} \right)$$

Taking real and imaginary parts:

$$\int_{-\infty}^{\infty} \cos(\lambda x) \frac{P(x)}{Q(x)} dx = \operatorname{Re} I \quad \text{and} \quad \int_{-\infty}^{\infty} \sin(\lambda x) \frac{P(x)}{Q(x)} dx = \operatorname{Im} I$$

Derivation (optional)

The above integral is a special case of the more general type

$$J = \int_{-\infty}^{\infty} \exp(i\lambda x) f(x) dx$$

where $f(z)$ is analytic except at most at a finite number of isolated points, and

- a. $f(z)$ has no poles on the real axis
- b. $f(z) \rightarrow 0$ as $|z| \rightarrow \infty$ for all z with $\operatorname{Im} z \geq 0$
- c. $\lambda > 0$

To evaluate J , we first integrate $\exp(i\lambda z) f(z)$ around a closed contour, C , which runs from $-R$ to R along the real axis, and then from R back to $-R$ along the semi-circle, C_R , of radius R and centre at the origin, in the upper half-plane:

$$\oint_C \exp(i\lambda z) f(z) dz = \int_{-R}^R \exp(i\lambda x) f(x) dx + \int_{C_R} \exp(i\lambda z) f(z) dz$$

Now we take $R \rightarrow \infty$ and apply the residue theorem:

$$\begin{aligned} \oint_C \exp(i\lambda z) f(z) dz &= 2\pi i \sum \left(\begin{array}{c} \text{residues of } [\exp(i\lambda z) f(z)] \\ \text{in upper half-plane} \end{array} \right) \\ &= \int_{-\infty}^{\infty} \exp(i\lambda x) f(x) dx + \lim_{R \rightarrow \infty} \int_{C_R} \exp(i\lambda z) f(z) dz \end{aligned}$$

Next we show that the integral, call it J_R , over C_R vanishes as $R \rightarrow \infty$ provided conditions **b.** and **c.** are satisfied. We have

$$|J_R| = \left| \int_{C_R} \exp(i\lambda z) f(z) dz \right| \leq \int_{C_R} |\exp(i\lambda z)| |f(z)| |dz| \leq M_R \int_{C_R} |\exp(i\lambda z)| |dz|$$

where M_R is the maximum of $|f(z)|$ on C_R .

On C_R : $z = R \exp(i\theta) = R(\cos \theta + i \sin \theta)$, $0 \leq \theta \leq \pi$ and $|dz| = |iR \exp(i\theta) d\theta| = R d\theta$. Also

$$\begin{aligned} \exp(i\lambda z) &= \exp[i\lambda R(\cos \theta + i \sin \theta)] = \exp(i\lambda R \cos \theta) \exp(-\lambda R \sin \theta) \\ &\Rightarrow |\exp(i\lambda z)| = \exp(-\lambda R \sin \theta) \end{aligned}$$

Then from the last inequality above:

$$\begin{aligned} |J_R| &\leq M_R \int_{C_R} |\exp(i\lambda z)| |dz| = M_R \int_0^\pi \exp(-\lambda R \sin \theta) R d\theta \\ &= 2M_R \int_0^{\pi/2} \exp(-\lambda R \sin \theta) R d\theta \end{aligned}$$

where for the last step we have used the fact that $\sin \theta$ is even about $\theta = \pi/2$. Now, for $0 \leq \theta \leq \pi/2$, $\sin \theta \geq 2\theta/\pi$ (easiest to see by sketching the functions $\sin \theta$ and $2\theta/\pi$). Then

$$\begin{aligned} |J_R| &\leq 2M_R \int_0^{\pi/2} \exp(-\lambda R \sin \theta) R d\theta \\ &< 2M_R \int_0^{\pi/2} \exp(-2\lambda R \theta/\pi) R d\theta \\ &< 2M_R \int_0^\infty \exp(-2\lambda R \theta/\pi) R d\theta \\ &= M_R \frac{\pi}{\lambda} \int_0^\infty \exp(-y) dy = M_R \frac{\pi}{\lambda} \end{aligned}$$

By condition **b.** $M_R \rightarrow 0$ as $R \rightarrow \infty$. Hence $|J_R| \rightarrow 0$ as $R \rightarrow \infty$.

This leaves

$$\int_{-\infty}^{\infty} \exp(i\lambda x) f(x) dx = 2\pi i \sum \left(\begin{array}{c} \text{residues of } [\exp(i\lambda z) f(z)] \\ \text{in upper half-plane} \end{array} \right)$$

To obtain our starting integral I , let $f(z) = \frac{P(z)}{Q(z)}$. Condition **ii.** ensures condition **b.** in the general case above.

Example

$$I = \int_{-\infty}^{\infty} \frac{\cos(ax)}{x^2 + b^2} dx \quad a, b \text{ real}$$

I does not depend on the signs of a and b : choose $a > 0$, $b > 0$ and let $\lambda = a$. Conditions **i** - **iii**. are satisfied.

We need the residues of

$$f(z) = \frac{\exp(iaz)}{z^2 + b^2} = \frac{\exp(iaz)}{(z + bi)(z - bi)}$$

There are simple poles at $z = \pm bi$. Only $z = bi$ is in the upper half-plane.

The residue at $z = bi$ is

$$r = \lim_{z \rightarrow bi} [(z - bi) f(z)] = \lim_{z \rightarrow bi} \frac{\exp(iaz)}{z + bi} = \frac{\exp(-ab)}{2bi}$$

Therefore

$$\int_{-\infty}^{\infty} \frac{\exp(iax)}{x^2 + b^2} dx = 2\pi i \frac{\exp(-ab)}{2bi} = \pi \frac{\exp(-ab)}{b}$$

Taking the real part:

$$\int_{-\infty}^{\infty} \frac{\cos(ax)}{x^2 + b^2} dx = \pi \frac{\exp(-ab)}{b}$$

Taking the imaginary part:

$$\int_{-\infty}^{\infty} \frac{\sin(ax)}{x^2 + b^2} dx = 0$$

as expected, since the integrand is an odd function.

18 Real integrals IV

For integrals of types II and III we can relax condition **i.** (no poles on the real axis) if the poles are of order 1.

Derivation (optional)

The closed contour C considered earlier is now modified by enclosing each pole on the real axis by a semi-circle, C_ϵ , of radius ϵ . The semi-circle lies in the upper half-plane, so that the poles on the real axis are left outside C .

Taking only one pole at x_0 as an example:

$$\oint_C = \int_{C_R} + \int_{-R}^{x_0-\epsilon} + \int_{C_\epsilon} + \int_{x_0+\epsilon}^R$$

We now take two limits: $R \rightarrow \infty$ (as before) and $\epsilon \rightarrow 0$.

Under the remaining conditions for integrals of type II, III, the integral over C_R again vanishes as $R \rightarrow \infty$.

We can show further that for $f(z)$ with a simple pole on the real axis

$$\lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} f(z) dz = -\pi i \text{ (residue of } f(z) \text{ on real axis)}$$

Furthermore

$$\lim_{R \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \left[\int_{-R}^{x_0-\epsilon} + \int_{x_0+\epsilon}^R \right] f(x) dx = \text{p.v.} \int_{-\infty}^{\infty} f(x) dx$$

where p.v. denotes the so-called principal value of the integral.

This leaves

$$\begin{aligned} & \text{p.v.} \int_{-\infty}^{\infty} f(x) dx \\ &= 2\pi i \sum \left(\begin{array}{c} \text{residues of } f(z) \\ \text{in upper half-plane} \end{array} \right) \\ & \quad + \pi i \sum \left(\begin{array}{c} \text{residues of } f(z) \\ \text{on real axis} \end{array} \right) \end{aligned}$$

Example 1

$$I = \int_{-\infty}^{\infty} \frac{\sin x}{x} dx$$

Consider

$$f(z) = \frac{\exp(iz)}{z}$$

which has a simple pole at $z = 0$ on the real axis, and no other poles.

The residue is

$$r = \lim_{z \rightarrow 0} z \frac{\exp(iz)}{z} = 1$$

Therefore

$$\text{p.v.} \int_{-\infty}^{\infty} \frac{\exp(ix)}{x} dx = \pi i r = \pi i$$

Taking real and imaginary parts:

$$\text{p.v.} \int_{-\infty}^{\infty} \frac{\cos x}{x} dx = 0$$

and

$$\text{p.v.} \int_{-\infty}^{\infty} \frac{\sin x}{x} dx = \pi$$

The first integral is zero because the integrand is an odd function.

In the second integral, we can now remove p.v., as the integrand in this case has a removable singularity:

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} dx = \pi$$

This is a well-known result.

Example 2

$$I = \int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx$$

We write the integral as

$$I = \frac{1}{2} \int_{-\infty}^{\infty} \frac{1 - \cos(2x)}{x^2} dx$$

Consider

$$f(z) = \frac{1 - \exp(2iz)}{z^2}$$

which has a simple pole at $z = 0$.

The residue is

$$\begin{aligned} r &= \lim_{z \rightarrow 0} z \frac{1 - \exp(2iz)}{z^2} \\ &= \lim_{z \rightarrow 0} \frac{1 - \exp(2iz)}{z} \\ &= \lim_{z \rightarrow 0} \frac{1 - (1 + 2iz - 2z^2 + \dots)}{z} \\ &= -2i \end{aligned}$$

Therefore

$$\text{p.v.} \int_{-\infty}^{\infty} \frac{1 - \exp(2ix)}{x^2} dx = \pi i r = 2\pi$$

Taking the real part:

$$\int_{-\infty}^{\infty} \frac{1 - \cos(2x)}{x^2} dx = 2\pi$$

Therefore

$$I = \int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx = \pi$$

This is another well-known result. (This and the previous example are related - can you see how?)